

■ A.5.2 Properties of Polyhedra

This section gives part of a database on properties of polyhedra. It is a common thing to use symbols like `Tetrahedron` to represent mathematical objects, and then symbolically to define properties such as `Area[Tetrahedron]`. Notice the use of `^=` to associate definitions with objects like `Tetrahedron`, rather than properties like `Area`.

```
(* Dodecahedron *)
Vertices[Dodecahedron] ^= 20
Edges[Dodecahedron] ^= 30
Faces[Dodecahedron] ^= 12
Area[Dodecahedron] ^= Sqrt[25+10 Sqrt[5]]/4
Inscribed[Dodecahedron] ^= Sqrt[250+110 Sqrt[5]]/20
Circumscribed[Dodecahedron] ^= (Sqrt[15]+Sqrt[3])/4
Volume[Dodecahedron] ^= (15 + 7 Sqrt[5])/4
Dual[Dodecahedron] ^= Icosahedron
```

Properties of polyhedra.